

9 (a) State what is meant by radioactive decay.

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..... [2]

(b) An isotope X is radioactive and decays to a stable product with a half-life of 12.0 minutes.

A sample of X has an activity of  $5.80 \times 10^{14}$  Bq at time  $t = 0$ .

(i) Calculate the decay constant of X. Give a unit with your answer.

decay constant = ..... unit ..... [2]

(ii) Calculate the activity of the sample at  $t = 20.0$  minutes.

activity = ..... Bq [2]

- (iii) On Figure 9.1, sketch the variation of the activity  $A$  of the sample with  $t$  between  $t = 0$  and  $t = 20.0$  minutes.

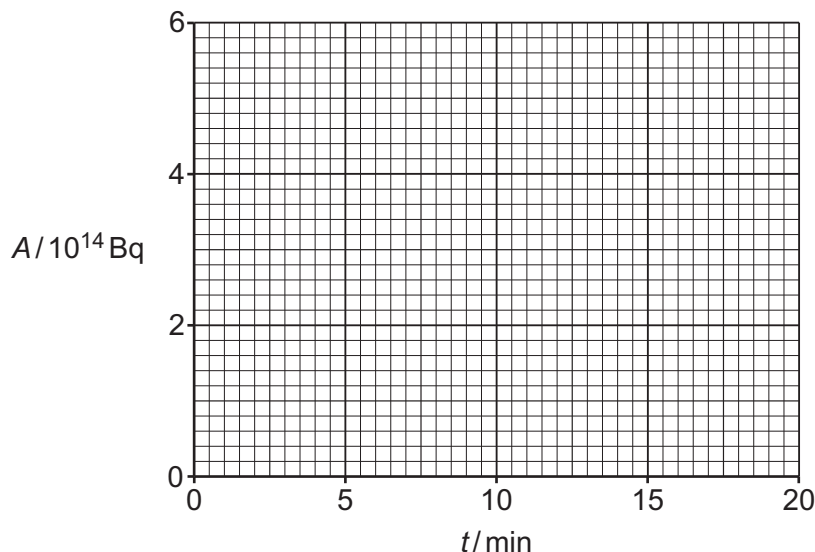


Figure 9.1

[3]

- (c) A scientist uses a radiation detector to determine the measured count rate of radiation emitted by the sample in **9(b)**.

Explain why the measured count rate at any given time is less than the activity of the sample.

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..... [3]

[Total: 12]